Supplemental Materials

This document provides supplemental materials for the manuscript entitled "Distributed Coverage Control on Poriferous Surface via Poly-Annulus Conformal Mapping" by Xun Feng, Chao Zhai, and it is composed of two parts. The first part presents some theoretical results, and it includes some key definitions and the derivation process of the mapping. The second part provides simulation results on the poriferous surface and relevant discussions, which include the analysis of computational cost, stability, and robustness. It is worth noting that the notations for theorems, lemmas, propositions, and citations in the supplementary material are consistent with those in the main text.

S-I Theoretical Results

A. Computational Complexity of the Partition Operator $\mathcal{P}$

The topological decomposition of the workspace S via the operator $\mathcal{P}$ is executed on a discrete triangular mesh, which is intrinsically modeled as an undirected graph $\mathcal{G} = (V, E)$. To efficiently construct the proximity partition, the algorithm initializes a min-heap priority queue with the vertices of the n inner boundaries ∂O_k. A multi-source Dijkstra algorithm is deployed to propagate the discrete geodesic distance field globally in a single traversal, incurring a time complexity of $\mathcal{O}(|E| \log |V|)$ [29]. Subsequently, to execute the branch cuts γ_k that annihilate local homology generators, localized shortest-path searches are performed within each disjoint proximity subregion (V_k, E_k). Because these subregions are mutually exclusive ($\sum_{k=1}^{n} |V_k| \leq |V|$ and $\sum_{k=1}^{n} |E_k| \leq |E|$), the aggregated complexity for all necessary branch cuts is strictly bounded by $\sum_{k=1}^{n} \mathcal{O}(|E_k| \log |V_k|) \leq \mathcal{O}(|E| \log |V|)$. Crucially, since the 2-D manifold S is discretized as a triangular surface mesh, its topological structure is governed by the Euler characteristic $\chi(S) = |V| - |E| + |F| = 1 - n$. Given that each triangular face is bounded by three edges ($3|F| \leq 2|E|$), the number of edges inherently satisfies the sparsity constraint $|E| \leq 3|V| - 3(1 - n)$, which establishes the asymptotic structural bound $|E| = \mathcal{O}(|V|)$. Substituting this sparsity property into the sequential graph traversals, the overall computational complexity of the partition operator $\mathcal{P}$ collapses to

$$\mathcal{O}(|V| \log |V|). \tag{S-1}$$

By replacing the iterative numerical approximations of continuous Eikonal PDE solvers with this discrete graph-theoretic formulation [8], the operator achieves exact topological decomposition without numerical diffusion, scaling efficiently with the mesh resolution.

B. Partial Welding Example

To illustrate the mathematical mechanism of partial welding, we present an intuitive example. Reconstruct the topological connectivity of a slit domain $\mathcal{D} = \mathbb{C} \setminus [-i, i]$ by welding its separated boundaries. Let the slit $[-i, i]$ be regarded as two distinct boundary segments: $\gamma^+ = \{iy + \epsilon \mid y \in [-1, 1]\}$, and $\gamma^- = \{iy - \epsilon \mid y \in [-1, 1]\}$, where $\epsilon \to 0^+$. The welding mapping $T : \mathcal{D} \to \Omega$ is defined by the elementary function $w = T(z) = \sqrt{z^2 + 1}$. This mapping transforms the slit domain $\mathcal{D}$ into the $\Omega = \mathbb{C} \setminus (-\infty, 0]$. Crucially, it enforces the boundary identification condition

$$T(iy^+) = T(iy^-) = \sqrt{1 - y^2}, \quad \forall y \in [-1, 1]. \tag{S-2}$$

This implies that the disjoint banks γ^+ and γ^- are mapped to the exact same locus in Ω, thereby welding the cut back together. In our proposed framework, the optimization of Θ_{uv} serves as the numerical inverse operator of such welding. It identifies the boundary correspondence

$$h_u(\partial S_u) \cong \Theta_{uv}(h_v(\partial S_v)), \tag{S-3}$$

which restores the continuous topology of poly-annulus S from the decomposed local disks.

C. Geometric Computation of Safe Boundary

This appendix details the numerical generation of the partition bar $\hat{\Gamma}_i$ in Ξ, consisting of linear segments and circular arcs. For intersection identification, the i-th nominal partition bar is parameterized as a vector $\hat{\mathbf{x}}(\hat{\xi}) = \hat{\xi}\hat{\mathbf{b}}_i$, where $\hat{\xi} \in [0, 1]$. The k-th obstacle is defined by its center $\hat{o}_k$ and radius $\hat{r}_k$. The intersection points are determined by the Euclidean constraint $\|\hat{\mathbf{x}}(\hat{\xi}) - \hat{o}_k\|^2 = \hat{r}_k^2$, which yields the quadratic equation

$$|\hat{\mathbf{b}}_i|^2 \hat{\xi}^2 - 2(\hat{\mathbf{b}}_i \cdot \hat{o}_k)\hat{\xi} + (|\hat{o}_k|^2 - \hat{r}_k^2) = 0. \tag{S-4}$$

Solving for $\hat{\xi}$ gives the entry and exit parameters $\hat{\xi}_{in} < \hat{\xi}_{out}$, with corresponding coordinates $\hat{p}_{in} = \hat{\xi}_{in}\hat{\mathbf{b}}_i$ and $\hat{p}_{out} = \hat{\xi}_{out}\hat{\mathbf{b}}_i$. To bypass the obstacle, a circular arc is generated on the buffer circle with radius $\hat{r}_{k,i} = \hat{r}_k(1 + i\beta)$. The entry and exit angles relative to $\hat{o}_k$ are computed as $\hat{\psi}_{in} = \text{atan}(\hat{p}_{in} - \hat{o}_k)$ and $\hat{\psi}_{out} = \text{atan}(\hat{p}_{out} - \hat{o}_k)$. The safe arc $\hat{A}_{k,i}$ is discretized via

$$\hat{\mathbf{x}}_{arc}(u) = \hat{o}_k + \hat{r}_{k,i} \begin{bmatrix} \cos(\hat{\psi}_{in} + u\frac{\Delta\tilde{\hat{\psi}}}{U-1}) \\ \sin(\hat{\psi}_{in} + u\frac{\Delta\tilde{\hat{\psi}}}{U-1}) \end{bmatrix}, \tag{S-5}$$

where $u \in \{0, \ldots, U-1\}$. This ensures $\hat{\Gamma}_i$ is piecewise smooth and strictly disjoint from obstacles.

D. Spectral Properties of $\mathcal{L}_{\mathcal{G}}(\hat{\omega})$

$$[\mathcal{L}_{\mathcal{G}}]_{ij} = \begin{cases} \hat{\omega}_i + \hat{\omega}_{i+1}, & \text{if } i = j \\ -\hat{\omega}_{\max(i,j)}, & \text{if } |i - j| = 1 \\ -\hat{\omega}_1, & \text{if } \{i, j\} = \{1, N\} \\ 0, & \text{otherwise} \end{cases} \tag{S-6}$$

For the weighted Laplacian $\mathcal{L}_{\mathcal{G}}(\hat{\omega})$ in (S-6), under $\hat{\omega}_k \geq \underline{\omega} > 0, \forall k \in \mathcal{N}$, the properties hold

1) $\mathcal{L}_{\mathcal{G}} = \mathcal{L}_{\mathcal{G}}^T$ and $\mathcal{L}_{\mathcal{G}} \succeq 0$.
2) $\ker(\mathcal{L}_{\mathcal{G}}) = \text{span}(\mathbf{1}_N)$, implying $\sum_{i=1}^{N} \dot{\hat{m}}_i = 0$.
3) $\lambda_2(\mathcal{L}_{\mathcal{G}}) \geq \underline{\omega}\lambda_2(\mathcal{L}_S) = 2\underline{\omega}(1 - \cos(2\pi/N)) > 0$.

Proof. Positive Semi-definiteness: The symmetry follows from the undirected adjacency in $\mathcal{G}_N$. For any $x \in \mathbb{R}^N$, the

quadratic form is $x^T\mathcal{L}_\mathcal{G}x = \sum_{k=1}^{N}\hat{\omega}_k(x_k - x_{k-1})^2 \geq 0$, where $x_0 \equiv x_N$. Then, $\mathcal{L}_\mathcal{G} \succeq 0$. Null Space: $x^T\mathcal{L}_\mathcal{G}x = 0 \iff x_k = x_{k-1}$ for all k. Hence, $\ker(\mathcal{L}_\mathcal{G}) = \text{span}(\mathbf{1}_N)$. Connectivity: Since $\underline{\omega} > 0$, the cycle graph $\mathcal{G}_N$ is connected. By the Courant-Fischer Theorem [37, Th. 4.2.2], the second smallest eigenvalue satisfies $\lambda_2(\mathcal{L}_\mathcal{G}) \geq \underline{\omega}\lambda_2(\mathcal{L}_S) > 0$, which establishes the exponential stability on the subspace $\mathbf{1}_N^\perp$. □

E. Theoretical Summary

The theoretical analysis constitutes a rigorous chain of logical deductions, which guarantees the system's reliability across geometric, dynamic, and computational implementations. First, Proposition III.1 establishes the topological conjugacy between the poly-annulus S and the n-holed disk Ξ via the conformal mapping τ. Building on this equivalent space, Proposition IV.1 and Lemma IV.1 confirm the strict topological safety of the modified sectorial partitions, ensuring collision avoidance while rigidly bounding the workload deviation induced by the sequence factor β. Regarding control feasibility, Lemma IV.2 proves the regularity of the pull-back Riemannian metric, which asymptotically recovers a static structure. Governed by this metric, Proposition IV.2 guarantees the non-degeneracy and forward invariance of the feasible set $\mathcal{F}_i$, rendering the Riemannian gradient flow globally well-defined. Subsequently, Proposition IV.3 ensures the existence and reachability of the optimal coverage configurations.

The stability analysis then reveals a robust cascaded convergence structure. Specifically, Theorem IV.1 rigorously proves that the virtual partition dynamics exhibit exponential input-to-state stability (ISS) against geometric perturbations. Coordinated with these robust partitions, Theorem IV.2 guarantees that the multi-agent system asymptotically converges to the optimal Riemannian centroids with local exponential rates. Furthermore, Theorem IV.3 bridges the gap between continuous theory and discrete implementation, validating that the numerical outputs strictly approximate the theoretical optimum within an arbitrarily small tolerance regulated by the precision parameters $(\Delta t, K^*, \beta)$. Finally, Theorem IV.4 establishes the topological universality of the framework, demonstrating that the gradient flow remains valid for the maximal class of Genus-0 manifolds, regardless of metric distortion and boundary complexity.

S-II Simulations

This section presents the selection of algorithm parameters and provides an analysis of the proposed framework, covering computational complexity, scalability, and robustness.

A. Parameter Design

The quantization parameter K^* and agent gain k_p are tuned to ensure that agents navigate the poly-annulus S toward their respective optimal centroids, while strictly adhering to the Riemannian metric constraints. The partition gain $k_{\hat{\psi}}$ is determined by the spectral properties of the communication topology to guarantee the exponential decay of workload imbalances across the multi-agent system. To maintain topological connectivity in the presence of holes, the sequence factor β is introduced to ensure that the obstacle-avoiding partition bars remain disjoint and strictly separated from the boundary components in Ξ. Finally, the convergence threshold T_ϵ is calibrated to define the steady-state precision of the coverage mission, ensuring that the terminal configuration satisfies the quasi-optimal workload distribution.

B. Computational Complexity

The computational execution of the proposed framework is structurally decoupled into a one-time topological initialization phase and an iterative coverage control phase.

1) **Topological Mapping (Algorithm 1):** Executed exactly once during the initialization. Solving the Laplace equation for the conformal map τ on S with $|V|$ vertices across M subdomains distributedly incurs a one-time complexity of $\mathcal{O}((|V| \cdot M^{-1})^{\frac{3}{2}})$.
2) **Sectorial Partition (Algorithm 2):** Executed iteratively. The linear intersection checks between the N partition bars and n obstacles yield a complexity of $\mathcal{O}(n)$ per control cycle.
3) **Coverage Control (Algorithm 3):** Executed iteratively. Each optimization iteration involves $\mathcal{O}(D_\mathcal{G})$ communication rounds, where $D_\mathcal{G}$ is the graph diameter.

Consequently, while the initial mapping requires $\mathcal{O}((|V| \cdot M^{-1})^{\frac{3}{2}})$ operations, the computational burden per control cycle significantly collapses to $\mathcal{O}(n + K^* \cdot D_\mathcal{G})$. As shown in Table I, the average execution time during the iterative phase is $t_{avg} \approx 0.02\,\text{s}$ per agent, which fully supports high-frequency real-time deployment.

C. Scalability Analysis

The poriferous geometry of S imposes a theoretical limit on the agent population N to maintain topological disjointness and input-to-state stability. Given a sequence factor β, the maximal agent capacity N_{max} is determined by

$$N_{max} \leq \min\left\{\frac{(\hat{r}_{max}\beta)^{-1} \cdot \hat{d}_{min}}{2}, \gamma\epsilon_{max} \cdot (C_{\hat{\delta}}\beta)^{-1}\right\},$$

where $\hat{d}_{min}$ is the minimum obstacle separation, $\hat{r}_{max}$ is the maximum obstacle radius, and ϵ_{max} is the allowable workload error. The first term ensures the non-overlapping of buffer zones ($\hat{r}_k(1 + N\beta) - \hat{r}_k \leq 0.5\hat{d}_{min}$), while the second term guarantees the ultimate workload error $\|\mathbf{e}(\infty)\| \leq C_{\hat{\delta}}\beta \cdot \gamma^{-1}$ remains within ϵ_{max}, where $C_{\hat{\delta}}$ is the geometric sensitivity constant defined in Lemma IV.1. This bound defines the fundamental trade-off between coverage precision β and the swarm scale N on the poly-annulus S.

D. Robustness Analysis

Consider a hardware contingency at $t = T_f$ where a subset of agents $\mathcal{N}_\mathcal{F}$ ceases operation, resulting in an active set $\mathcal{A}$ with size $N_\mathcal{A} = N - N_\mathcal{F}$. The new equilibrium workload becomes $\bar{m}_\mathcal{A} = M_{total} \cdot N_\mathcal{A}^{-1}$. Define the error vector $\mathbf{e} \in \mathbb{R}^{N_\mathcal{A}}$ with components $e_i = \hat{m}_i - \bar{m}_\mathcal{A}$, satisfying $\mathbf{e} \perp \mathbf{1}_{N_\mathcal{A}}$. Under the reduced topology, the error dynamics in Ξ follow

the discrete Laplacian flow $\dot{\mathbf{e}} = -k_{\hat{\psi}}\mathcal{L}_{\mathcal{G}}(\hat{\omega})\mathbf{e}$, where $\mathcal{L}_{\mathcal{G}}(\hat{\omega})$ is the weighted Laplacian. For $V = \|\mathbf{e}\|^2/2$, its derivative satisfies

$$\dot{V} = -k_{\hat{\psi}}\mathbf{e}^T\mathcal{L}_{\mathcal{G}}(\hat{\omega})\mathbf{e} \leq -k_{\hat{\psi}} \cdot \underline{\omega} \cdot \lambda_2(\mathcal{L}_{S,N_{\mathcal{A}}}) \cdot \|\mathbf{e}\|^2, \quad \text{(S-7)}$$

where $\underline{\omega} > 0$ is the minimum marginal density. The exponential convergence rate is $\gamma_{\mathcal{A}} = 2k_{\hat{\psi}} \cdot \underline{\omega} \cdot (1 - \cos(2\pi \cdot N_{\mathcal{A}}^{-1}))$. Since $N_{\mathcal{A}} < N$, the spectral property of the cycle graph inherently ensures $\gamma_{\mathcal{A}} > \gamma_{\mathcal{N}}$. The re-balancing speed automatically increases as the swarm size reduces, ensuring that the error $\|\mathbf{e}(t)\| \leq \|\mathbf{e}(T_f)\| \exp(-\gamma_{\mathcal{A}}(t - T_f))$ swiftly decays. This intrinsic self-healing property allows the remaining agents to reach new Riemannian centroids without requiring global reconfiguration of the conformal map τ.